

In a recent paper, Lagarias, Prabhu & Reeds [347] discussed the configuration space of all d -step configurations for a fixed d , analyze its structure, and relate the d -step problem to certain factorization problems for matrices. They also suggested that there might in fact be at least 2^{d-1} paths of length d between the complementary vertices of any d -dimensional Dantzig figure. However, Klee & Holt [281, 282] have now shown that this is true for $d \leq 4$, but false for all $d > 4$.

Our treatment of the reconstruction of polytopes from their graphs owes heavy thanks (thefts) to the paper [299] of Gil Kalai, as indicated there. We might repeat here that the graph of a polytope carries important information, but by far not all the relevant information about the structure of a polytope. One aspect of this is the fact that the graph does not determine the dimension of a general polytope; see also Exercise 3.4. Another one is that there are far fewer different polytope graphs for various parameters than there are different polytopes. In fact, Perles proved that the number of nonisomorphic graphs of d -polytopes with $d+k$ vertices is bounded by a function of k (independent from d !). The proof for that — see Kalai [307] — uses only some lemmas about finite set systems. In contrast to Perles' result one can easily see (for example with the methods of Section 6.5) that the number of different d -polytopes with $d+2$ vertices (i.e., $k=2$) is not bounded.

Problems and Exercises

- 3.0 (**Stellar subdivisions** [204]). Let F be a facet of the d -polytope $P \subseteq \mathbb{R}^d$, and construct a point $\mathbf{y}_F \in \mathbb{R}^d$ beyond F . The polytope

$$\text{st}(P, F) := \text{conv}(P \cup \{\mathbf{y}_F\})$$

is the *stellar subdivision* of P at F .

- (i) Describe the faces of $\text{st}(P, F)$ in terms of faces of P . Conclude that the combinatorial type of $\text{st}(P, F)$ does not depend on the precise position of $\mathbf{y}_F$.
- (ii) Show that if P is simplicial, then so is $\text{st}(P, F)$. In this case, count the number of k -faces of $\text{st}(P, F)$ in terms of the numbers $f_i(P)$ of i -faces of P .
- (iii) Describe the operation that is “polar” to stellar subdivision, given by

$$\text{st}^\Delta(P, \mathbf{v}) := (\text{st}(P^\Delta, \mathbf{v}^\diamond))^\Delta,$$

for any vertex $\mathbf{v}$ of P .

- 3.1 For a vertex $\mathbf{v}$ of the d -polytope P , and $k \geq 1$, construct the cones generated by all vertices of P of distance k from $\mathbf{v}$:

$$C_k := \text{cone}\{\mathbf{w} - \mathbf{v} : \mathbf{w} \in \text{vert}(P), \delta(\mathbf{v}, \mathbf{w}) = k\}.$$

Prove the “nested cones theorem” of Hochstättler [275]:

$$C_1 \supseteq C_2 \supseteq C_3 \supseteq \dots$$

- 3.2 If P has dimension at least 4, then the graph $G(P)$ is not planar. In fact, show that it contains a subdivision of the complete graph K_{d+1} (Grünbaum [252, pp. 200, 214]).
- 3.3 If $n > d \geq 4$, then the graph of the cyclic polytope $C_d(n)$ is complete, $G(C_d(n)) = K_n$. Give a direct proof: for each edge, construct an explicit linear function that is maximized by this edge.
- 3.4 A d -polytope P is called *dimensionally ambiguous* if there is a polytope Q of a different dimension $\dim(Q) \neq \dim(P)$ which has an isomorphic graph, $G(P) \cong G(Q)$.
- (i) Show that the d -simplex is dimensionally ambiguous for $d \geq 5$, but not for $d \leq 4$.
 - (ii) Show that 3-polytopes, and simple 4-polytopes, cannot be dimensionally ambiguous. (Hint: Use Exercise 3.2!)
 - (iii) Show that if P is a 0/1-polytope whose graph is isomorphic to $G(C_d)$, then P is affinely isomorphic to C_d . (Compare to Exercise 2.5!)
 - (iv) Show that the d -cubes are dimensionally ambiguous. In particular, describe a 4-polytope whose graph is isomorphic to $G(C_5)$! (Suitable polytopes can be constructed directly, or can be identified within Blind & Blind’s [107] classification of all cubical d -polytopes with 2^{d+1} vertices. Indeed there are cubical 4-polytopes with the graph of the n -cube, for any $n \geq 4$. More generally, “neighborly cubical polytopes exist!” — see Babson, Billera & Chan [33] and Joswig & Ziegler [294].)
- 3.5 If $P = P(A, \mathbf{1})$ is an irredundant description, show that for small enough $\lambda > 0$ the polytope

$$P' := P(A, \mathbf{1}^{(\lambda)}),$$

with $\mathbf{1}^{(\lambda)}_i = 1 + \lambda^i$ as in Lemma 3.2, is a simple polytope whose facets are in natural bijection with the facets of P .

Furthermore, show that then $\delta(G(P')) \geq \delta(G(P))$: thus it is sufficient to prove the Hirsch conjecture for simple polytopes.

- 3.6 If P is a pointed polyhedron in $\mathbb{R}^3$, show that the graph of all bounded edges is connected. Show that it is not 2-connected in general. What about higher dimensions?
- 3.7 Let $P \subseteq \mathbb{R}^d$ be a d -polytope with $2d$ facets, such that the facets containing $\mathbf{v} = \mathbf{0}$ determine the positive orthant $\mathbf{u} \geq \mathbf{0}$. Show that the facets of P can have $2d - 1$ facets each, if $d \geq 4$. (This is why it is hard to use inductive arguments for the d -step conjecture.)

- 3.8 Prove the following theorem by Kleinschmidt & Onn [336]: if P is a d -polytope whose vertex set is contained in $\{0, 1, \dots, k\}^d$, then the diameter of its graph is bounded by

$$\delta(G(P)) \leq kd.$$

Why can't this be used to get effective bounds for the diameters of d -polytopes? (See Problem 4.16* and Section 6.5(a) for answers.)

- 3.9 Consider the simplex algorithm, applied to a linear function c on a simple, d -dimensional polyhedron with at most n facets, such that there is a unique optimal vertex.

- (i)* The EDGE-RANDOM rule moves along random increasing edges, where at any given vertex the increasing edges leaving it are taken with equal probability. Can you give any subexponential upper bound (in n and d) for the expected number of steps of this rule on any linear programming problem? Is there a polynomial upper bound?
- (ii) Assume that we use the following RANDOM-FACET pivot rule to choose the increasing edge. That is, at the starting vertex v ,
- if $\text{up-degree}(v) = 0$, then STOP — the current vertex is optimal,
 - if $\text{up-degree}(v) = 1$, then take the unique increasing edge,
 - if $\text{up-degree}(v) > 1$, then take a random facet among facets that contain v , restrict the linear program to that facet, and solve the restriction by a recursive call to RANDOM-FACET.
- (This is what Kalai in [306] calls an “antipivot rule,” or the “bureaucratic” rule.)

Show that the maximal expected running time of this algorithm can be bounded by a function $E(d, n)$, which for $n > d$ satisfies the recursion

$$E(d, n) \leq \max \left\{ 1 + E(d, n-1), \right. \\ \left. E(d-1, n-1) + \frac{1}{d} \sum_{i=1}^{\max\{d, n-d\}} E(d, n-i) \right\}.$$

(These recursions are in fact not hard to see; the derivation of the asymptotics implied by this recursion, $E(n, d) \leq n^{4\sqrt{d}}$, is not that easy (i.e., difficult). See Matoušek, Sharir & Welzl [383] for a careful treatment of the asymptotic analysis.)

(Both rules are studied in Gärtner, Henk & Ziegler [219]. In particular, for special linear programming problems it is shown that there are starting vertices for which the expected number of steps is (nearly) quadratic. See also [306].)

- 3.10 (Basis version of linear programming).** Let $P = P(A, \mathbf{z}) \subseteq \mathbb{R}^d$, and let $\mathbf{c}\mathbf{x}$ be a linear function on $\mathbb{R}^d$. A subset of d of the inequalities of P , say $A'\mathbf{x} \leq \mathbf{z}'$, is a *basis* if A' has rank d (equivalently, $A'\mathbf{x} = \mathbf{z}'$ has a unique solution $\mathbf{x} \in \mathbb{R}^d$). A basis $A'\mathbf{x} \leq \mathbf{z}'$ is *feasible* if the unique solution of $A'\mathbf{x} = \mathbf{z}'$ satisfies $\mathbf{x} \in P$, and *dual feasible* if it maximizes $\mathbf{c}\mathbf{x}$ over $\{\mathbf{x} \in \mathbb{R}^d : A'\mathbf{x} \leq \mathbf{z}'\}$ (equivalently, if $\mathbf{0}$ maximizes $\mathbf{c}\mathbf{x}$ over $\{\mathbf{x} \in \mathbb{R}^d : A'\mathbf{x} \leq \mathbf{0}\}$).
- (i) If $\mathbf{v}$ is a vertex of $P \subseteq \mathbb{R}^d$, show (using Carathéodory's Theorem 1.15) that there is a feasible basis that has $\mathbf{v}$ as its solution. If P is simple, the basis is unique.
 - (ii) Show that all the feasible bases for a vertex of P are connected by sequences of single-element exchanges; that is, in every such exchange one inequality from the system $A'\mathbf{x} \leq \mathbf{z}'$ is replaced by a single different one from the big system $A\mathbf{x} \leq \mathbf{z}$.
 - (iii) If E is an edge of P adjacent to $\mathbf{v}$, show that there is a feasible basis for $\mathbf{v}$ such that all but one of the inequalities $A'\mathbf{x} \leq \mathbf{z}'$ are satisfied on E with equality.
 - (iv) Show that if a basis is both feasible and dual feasible, then it is an optimal solution for the program $\max \mathbf{c}\mathbf{x}, A\mathbf{x} \leq \mathbf{z}$.
 - (v) Use a Farkas lemma to prove that if P is a nonempty polytope, then there is a basis that is both feasible and dual feasible.
- 3.11*** For a d -dimensional polytope with n facets, what is the maximal number $M(d, n)$ of vertices in a monotone path?
- (This $M(d, n)$ is an upper bound on the largest number of steps in any simplex algorithm. It is known (see Klee & Minty [330]) that $M(d, 2d) \geq 2^d$, and roughly that $M(d, n) \geq \lfloor \frac{n}{d/2} \rfloor^{\lfloor d/2 \rfloor}$.
- One might think that the upper bound given by the upper bound theorem could be sharp — it is not; see Problem 8.41*. There is a suggestion by Klee & Minty [330, p. 175] that there could be a function γ such that $M(d, n) \leq \gamma(n - d)d$.)
- 3.12*** Does every simple 4-dimensional polytope have a Hamilton cycle?
- (This conjecture is due to Barnette; see [203, p. 158]. Some special cases are in Barnette & Rosenfeld [54].)
- 3.13*** Let P be a simple 4-polytope. Is it true that every connected planar 3-regular subgraph that does not separate $G(P)$ is the graph of a facet of P ? (Without the planarity condition, this had been a conjecture of Perles [299], which was disproved in [263].)
- 3.14** A k -path between distinct vertices $\mathbf{v}$ and $\mathbf{w}$ of a d -polytope P is a sequence of k -faces $F_1, \dots, F_m$ such that $\mathbf{v}$ is a vertex of F_1 , $\mathbf{w}$ is a vertex of F_m , and F_i and F_{i+1} are adjacent for $1 \leq i < m$ (i.e., their intersection is a $(k - 1)$ -face). Two k -paths are disjoint if they have no k -face in common.

- (i) Derive from Balinski's theorem that for $k = 1$ and for $k = d - 1$ there are d disjoint k -paths between any two vertices v and w .
- (ii) If P is a simplex, show that there are $\binom{d}{k}$ disjoint k -paths between any two vertices v and w .
- (iii)* Are there $\binom{d}{k}$ disjoint k -paths between any two vertices v and w of any d -dimensional polytope?

(The problem is due to Prabhu [446], from 1990; no progress, yet.)

3.15 A *complex* Δ is a collection of d -subsets of the set $[n] := \{1, 2, \dots, n\}$. (This defines an “abstract simplicial complex” in the sense of Section 8.5.) Two sets $F, G \in \Delta$ are *adjacent* if they differ only in one element. This defines a graph on the d -sets in Δ , and Δ is called *strongly connected* if this graph is connected. Even stronger, Δ is called *ultraconnected* if every nonempty subfamily of the form $\Delta_K := \{F \in \Delta : K \subseteq F\}$ is strongly connected.

- (i) Show that the complex which corresponds to a simplicial d -polytope with vertex set (identified with) $[n]$ is ultraconnected.
- (ii) If P is a simple d -dimensional polyhedron whose set of facets is labeled by $[n]$, then there is a d -set associated with every vertex. Show that the corresponding complex is ultraconnected.
- (iii) Every shellable simplicial complex is ultraconnected. (Shellability is an important combinatorial concept: see Section 8.1 for the definition.)

More generally, a pure simplicial complex is shellable if and only if it has an ordering $F_1, F_2, \dots, F_s$ of its facets such that the subcomplex $F_1 \cup F_2 \cup \dots \cup F_s$ is ultraconnected for all i .

- (iv) Let Δ be the complex of the 4-sets 1234, 2345, 1346, 5678, 2678, 1578, and all the 4-sets that have two elements from 1234 and two elements from 5678 but do not contain both 1 and 2 nor both 5 and 6. Show that the distance between 1234 and 5678 in Δ is 5. Show that Δ is ultraconnected.
- (v) Describe an ultraconnected complex of triangles (i.e., $d = 3$) on n vertices with diameter $n - 3$.
- (vi) Let Δ be an ultraconnected complex of triangles on n vertices. Show that between any two triangles there is a path of triangles which visit every vertex at most twice. Deduce that the diameter is at most $2n$.
- (vii)* Can you improve $2n$ to $1.999n$ (or at least to $2n - 1000$, $2n - 1$)? Can you find an ultraconnected collection of triangles on n vertices with diameter $> 1.001n$? (or at least $n + 100$, or even $n - 2$?)

(This combinatorial set-up for studying diameter questions is due to Larman [349] and to Kalai [305]. In particular, in [349] Larman showed that between every two vertices of an ultraconnected complex of d -sets on n vertices, there is a path that visits every vertex at

most 2^{d-1} times, and this implies a bound of $\Delta_u(d, n) \leq 2^{d-1}n$. (See also Klee & Kleinschmidt [327, Sect. 7].)

The unbounded 4-polyhedron with 8 facets by Klee & Walkup [331], which fails the Hirsch bound, yields the complex of part (iv).

Subexponential diameter bounds for ultraconnected complexes were found by Kalai, see [305, Sect. 4.1]. This exercise is also due to him. Part (vii)* demonstrates the unbelievable gap between the known lower and upper bounds. Note that by parts (i) and (ii), every upper bound that one can prove for the diameter of an ultraconnected complex of d -sets is automatically also valid for $\Delta_u(d, n)$.

- 3.16 Let P be a simple d -polytope such that every k -face of P has at most $2k$ facets.

- (i) Show that the diameter of P is bounded above by d .
- (ii) Moreover show for such polytopes that for every objective function and any starting point one can “reach the top” in d steps.

(This is from Kalai [305, Thm. 3], where it is proved that for every fixed $r \geq 2$, if every k -face of P has at most rk facets, then the diameter and the height of P are bounded by a polynomial in d .)

- 3.17 Given finite graphs G and H , we define that G is an *induced subgraph* of H if we can obtain a graph isomorphic to G by deleting a set S of vertices (and all edges adjacent to them) from H . We say that H is a *suspension* of G if additionally we require that the vertices of S are connected to all other vertices of H .

- (i) Show that every finite graph is an induced subgraph of the graph of a 4-polytope.
(If G has $n \geq 5$ vertices, then start with $C_4(n)$, and introduce an extra vertex beyond every edge that is missing in G .)
- (ii) For every finite graph there is some suspension which is the graph of a d -polytope, for some d .
(If G has $n \geq 5$ vertices, then start with $C_4(n)$, and introduce an extra dimension and two new vertices for every edge in G that is supposed to be missing.)
- (iii)* Does every finite graph have a suspension that is the graph of a 4-polytope? What about the case where G is the graph with n vertices but no edges?
- (iv) Give an example of a 4-connected graph on $n \geq 5$ vertices which is not the graph of a 4-polytope. Can you construct a 4-regular graph with these properties?

(Perles [438])

4

Steinitz' Theorem for 3-Polytopes

The combinatorial structure of 2-polytopes is not much of a mystery. For an illustrated journey into the wonderful world of 3-dimensional convex (and nonconvex) polyhedra, we direct the reader's attention to the conference volume [492]. See also Barnette's book [45] for a nice elementary discussion of combinatorics and graph theory related to convex 3-polytopes.

In this lecture we will establish the basic theory for 3-polytopes, by proving *Steinitz' theorem*. The basic version reads as follows.

Theorem 4.1 (Steinitz' theorem). [524, 527]

G is the graph of a 3-dimensional polytope if and only if it is simple, planar, and 3-connected.

Polytopal graphs are certainly *simple*: they have no loops or multiple edges. The graph $G(P)$ is planar for every 3-polytope (use radial projection to a sphere from an interior point, or a linear projection to a plane from a point beyond a facet). Also, $G(P)$ is 3-connected by Balinski's Theorem 3.14. Thus, the difficult part is the “if” part of the theorem: it requires that we show how, given a 3-connected planar graph, one can construct a 3-polytope.

